CS3101: Class Test (makeup)

Date: 21/11/2024 7 PM **Duration: 1 hour 30 mins** **Total marks: 25**

Instructions

- All programs should be compatible with GNU C compiler.
- There will be a plagiarism check. Copying from internet will also be considered as plagiarism.
- Late submissions will attract marks deduction.
- You are advised to compile the code before submitting them to WeLearn. If a code does not compile at our end, marks will be deducted for the problem.
- Each program should follow a strict naming convention: **QNo.c** (e.g. Q1.c, Q2a.c etc.). Programs not adhering to the convention will not be considered for evaluation.
- All codes (.c files) should be submitted in a single zipped folder (folder name containing your WeLearn ID e.g. 17MS001) to WeLearn.
- You should put appropriate comments in the code. Importantly, junctures in the code that involve critical decision-making must be accompanied with comments to enhance the readability of the same.

1. **(Marks: 7)** Write a C program to consider an array (no need to take user input), reverse it without using another array (the original array will be reversed), and print the reversed array.

2. **(Marks: 9)** In C programming language write a **recursive** function `displayPattern()`, that takes n, the number of rows in the pattern (and any other necessary value) as argument(s), to generate a pattern shown in Figure 1, such that each row is produced by each recursive call. You are NOT allowed to use global variables.

Figure 1: Pattern for Q2 (n = 5)

3. **(Marks: 9)** Consider an encryption scheme where a string "SHERLOCK" gets encrypted to "REHS*KCOL". Write a decryption function `decrypt()` that takes a string **S** (e.g. REHS*KCOL) as an argument (use pointers), and returns its decrypted form (here, SHERLOCK). For simplicity, you may assume that **S** has the desired format: <string>*<string>. You are NOT allowed to use global variables.